

Industry 4.0

Section Id :	64065364223
Section Number :	5
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	22
Number of Questions to be attempted :	22
Section Marks :	45
Display Number Panel :	Yes
Section Negative Marks :	0
Group All Questions :	No
Enable Mark as Answered Mark for Review and Clear Response :	No
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	640653134747
Question Shuffling Allowed :	No

Question Number : 106 Question Id : 640653906829 Question Type : MCQ Calculator : Yes
Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DEGREE LEVEL : INDUSTRY 4.0 (COMPUTER BASED EXAM)"

**ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?
CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.**

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

6406533052398. ✓ YES

6406533052399. ✗ NO

Sub-Section Number :	2
Sub-Section Id :	640653134748
Question Shuffling Allowed :	Yes

Question Number : 107 Question Id : 640653906830 Question Type : MCQ Calculator : Yes
Correct Marks : 1

Question Label : Multiple Choice Question

Match Column A with Column B (country of origin)

Column A	Column B
A1. 1 st revolution	B1. England
A2. 2 nd revolution	B2. USA
A3. 3 rd revolution	B3. USA/Japan
A4. 4 th revolution	B4. Germany

Options :

6406533052400. ✓ A1:B1, A2:B2, A3:B3, A4:B4

6406533052401. ✗ A1:B2, A2:B1, A3:B3, A4:B4

6406533052402. ✗ A1:B3, A2:B1, A3:B4, A4:B2

6406533052403. ✗ A1:B3, A2:B1, A3:B2, A4:B4

Sub-Section Number :

3

Sub-Section Id :

640653134749

Question Shuffling Allowed :

Yes

Question Number : 108 Question Id : 640653906832 Question Type : MCQ Calculator : Yes

Correct Marks : 1.5

Question Label : Multiple Choice Question

Geet Mobile manufactures mobile phones of various types. Their product portfolio comprises of 15 types of headsets. These headsets comprise of limited types of components. Only a few of the components needs customization for different headsets. Which facility layout will best suit for Geet Mobile's production?

Options :

6406533052408. ✗ Product layout

6406533052409. ✓ Group layout

6406533052410. ✗ Fixed position layout

6406533052411. ✗ All of these

Sub-Section Number :

4

Sub-Section Id :

640653134750

Question Shuffling Allowed :

Yes

Question Number : 109 Question Id : 640653906851 Question Type : MCQ Calculator : Yes

Correct Marks : 2

Question Label : Multiple Choice Question

Mahesh is shopping in a departmental store. He has a list of items which he prepared at home itself. He picked a few packets of tea as part of his list. He could see coffee packets and milk-powder packets in the neighbouring shelves. He picked a few milk-powder packets, however these are not part of his list. While standing in the queue for billing, he noticed Uncle Chips packets lying nearby and grabbed a couple of packets. Now map the scenarios with the items below:

Product	Scenario
P1) Tea : Coffee	S1) Complementary
P2) Tea: Milk-powder packet	S2) Substitution
P3) Uncle chips purchase	S3) Inventory dependant demand
	S4) Impulse buying

Options :

6406533052446. ✖ P1-S1, P2-S3, P3-S4

6406533052447. ✖ P1-S2, P2-S1, P3-S3

6406533052448. ✔ P1-S2, P2-S1, P3-S4

Question Number : 110 Question Id : 640653906854 Question Type : MCQ Calculator : Yes

Correct Marks : 2

Question Label : Multiple Choice Question

What is the name of the forecasting method represented by the following set of equations? The notations represent the usual meaning.

$$F_{t+1} = a_t + b_t$$

Where

$$a_t = \alpha D_t + (1 - \alpha)(a_{t-1} - b_{t-1});$$

$$b_t = \beta(a_t - a_{t-1}) + (1 - \beta)b_{t-1}$$

Options :

6406533052456. ✖ Exponential smoothing

6406533052457. ✖ Winter's model for seasonality

6406533052458. ✔ Holte's method for trend

6406533052459. ✖ None of these

Question Number : 111 Question Id : 640653906856 Question Type : MCQ Calculator : Yes

Correct Marks : 2

Question Label : Multiple Choice Question

Which of the following is not an example of digitization?

Options :

6406533052465. ✖ Creating the e-copy of a book using Google-Lens

6406533052466. ✔ Generating an automated daily sales report across the retail outlets in a state.

6406533052467. ✖ Converting a physical 2D sheet to CAD drawing

6406533052468. ✖ Converting physical invoice to e-copy

Question Number : 112 Question Id : 640653906857 Question Type : MCQ Calculator : Yes

Correct Marks : 2

Question Label : Multiple Choice Question

Select the correct form of internet from column B corresponding to the example in Column A.

Column A	Column B
A1. Smart metering	B1. Consumer
A2. Banking system	B2. Commercial
A3. Entertainment integrated devices	B3. Industrial
	B4. Private

Options :

6406533052469. ✖ A1: B3, A2: B2, A3: B4

6406533052470. ✔ A1: B3, A2: B2, A3: B1

6406533052471. ✖ A1: B2, A2: B3, A3: B1

Sub-Section Number :

5

Sub-Section Id :

640653134751

Question Shuffling Allowed :

Yes

Question Number : 113 Question Id : 640653906836 Question Type : MSQ Calculator : Yes

Correct Marks : 1 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which are the two primary tasks of time series forecasting?

Options :

6406533052418. ✔ Identifying pattern

6406533052419. ✖ Identify variables of interest

6406533052420. ✔ Discovering relationships amongst variables of interest

Question Number : 114 Question Id : 640653906837 Question Type : MSQ Calculator : Yes

Correct Marks : 1 Max. Selectable Options : 0

Question Label : Multiple Select Question

Select the correct statements from below:

Options :

6406533052421. ✖ Optimal solution in a facility location problem using Euclidean method will result in a region

6406533052422. ✔ Optimal solution in a facility location problem using Euclidean method rarely matches the center of gravity

6406533052423. ✔ Optimal solution in a facility location problem using Euclidean method will result in a point

Question Number : 115 Question Id : 640653906852 Question Type : MSQ Calculator : Yes

Correct Marks : 1 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following example(s) can be referred as seasonal pattern?

Options :

- 6406533052449. ✖ Economic recession
- 6406533052450. ✔ Increases Ceiling Fan sales during winters
- 6406533052451. ✔ Increased tourist footfall towards the end of the year

Sub-Section Number : 6
Sub-Section Id : 640653134752
Question Shuffling Allowed : Yes

Question Number : 116 Question Id : 640653906831 Question Type : MSQ Calculator : Yes
Correct Marks : 1.5 Max. Selectable Options : 0

Question Label : Multiple Select Question
Select the appropriate purposes of facility layout from the list below:

Options :

- 6406533052404. ✔ Minimize delay in material handling
- 6406533052405. ✔ Enhance sales as appropriate in manufacturing and service facilities.
- 6406533052406. ✔ Provide for good maintenance
- 6406533052407. ✖ Funding a new project

Question Number : 117 Question Id : 640653906835 Question Type : MSQ Calculator : Yes
Correct Marks : 1.5 Max. Selectable Options : 0

Question Label : Multiple Select Question
There are four demand points at A(1,2), B(2,3), C(3,5), D(4,1). The respective demand at these points are 7, 1, 3, 5. Select the optimal location based on cross-median approach.

Options :

- 6406533052414. ✔ (2,2)
- 6406533052415. ✔ (3,2)
- 6406533052416. ✔ (2.5,2)
- 6406533052417. ✖ (2,3)

Sub-Section Number : 7
Sub-Section Id : 640653134753
Question Shuffling Allowed : Yes

Question Number : 118 Question Id : 640653906838 Question Type : MSQ Calculator : Yes
Correct Marks : 2 Max. Selectable Options : 0

Question Label : Multiple Select Question
Select the demerits of outsourcing from the following list.

Options :

- 6406533052424. ✔ Reduced customer/supplier contact.
- 6406533052425. ✔ Loss of supply chain visibility.
- 6406533052426. ✔ Leakage of sensitive data and information.

6406533052427. ✓ Loss of internal capability and growth in third-party power.

Question Number : 119 Question Id : 640653906853 Question Type : MSQ Calculator : Yes
Correct Marks : 2 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following is/are considered in the Vehicle Routing Problem but not considered in TSP?

Options :

6406533052452. ✗ Subtour elimination

6406533052453. ✓ Any capacity restrictions

6406533052454. ✓ A specific starting point

6406533052455. ✗ Distance travelled restriction

Question Number : 120 Question Id : 640653906855 Question Type : MSQ Calculator : Yes
Correct Marks : 2 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following supply chain costs is/are affected by distribution network structure?

Options :

6406533052460. ✓ Inventory

6406533052461. ✓ Transportation

6406533052462. ✓ Facilities and handling

6406533052463. ✓ Information

6406533052464. ✗ Price

Sub-Section Number :

8

Sub-Section Id :

640653134754

Question Shuffling Allowed :

Yes

Question Number : 121 Question Id : 640653906833 Question Type : SA Calculator : None
Correct Marks : 2

Question Label : Short Answer Question

A factory works for two 8-hour shifts each day and produces a total of 24,000 electric bulbs using a set of workstations. There are 8 activities required to manufacture the bulb, such that the sum of all activity times is equal to 12 seconds. How many workstations are required to maintain the specified level of production (assuming that activities can be combined into those workstations in any order) (enter your answer in numeric value)?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

5

Sub-Section Number : 9
Sub-Section Id : 640653134755
Question Shuffling Allowed : Yes

Question Number : 122 Question Id : 640653906834 Question Type : SA Calculator : None
Correct Marks : 1.5

Question Label : Short Answer Question
Calculate the Euclidean distance between two machines located at the coordinates A(5,10) and B(2,6) in a shop floor.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Range
Text Areas : PlainText
Possible Answers :

4.8 to 5.3

Sub-Section Number : 10
Sub-Section Id : 640653134756
Question Shuffling Allowed : Yes

Question Number : 123 Question Id : 640653906861 Question Type : SA Calculator : None
Correct Marks : 2

Question Label : Short Answer Question
What is the full form of SAAS?

NOTE:Enter the exact answer without any space in the beginning or at the end.

Response Type : Alphanumeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Answers Case Sensitive : No
Text Areas : PlainText
Possible Answers :

Software as a service

Sub-Section Number : 11
Sub-Section Id : 640653134757
Question Shuffling Allowed : No

Question Id : 640653906858 Question Type : COMPREHENSION Sub Question Shuffling
Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Calculator : None

Question Numbers : (124 to 125)

Question Label : Comprehension

Suppose you need to accept reservation for a flight. There are total 100 seats in the aircraft. You have two classes of passengers - economy and business. One economy seat gets you 2,500 Rs of revenue while a business seat gets 7,500 Rs. From your experience, you know the demand for business travellers follows a uniform distribution in [10,20]. Then answer the given subquestions:

Sub questions

Question Number : 124 Question Id : 640653906859 Question Type : SA Calculator : None

Correct Marks : 1.5

Question Label : Short Answer Question

What is the Booking limit?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

17

Question Number : 125 Question Id : 640653906860 Question Type : SA Calculator : None

Correct Marks : 2

Question Label : Short Answer Question

What is the protection level?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

83

Sub-Section Number :

12

Sub-Section Id :

640653134758

Question Shuffling Allowed :

No

Question Id : 640653906843 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Calculator : None

Question Numbers : (126 to 129)

Question Label : Comprehension

Limat Ltd. wants to determine its order quantity for one of its components. The annual

consumption for the component is 80,000 units and its purchasing price is Rs. 50 per unit. The cost of placing an order is Rs. 1,200 and carrying cost per unit is 6% of the purchasing price.

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 126 Question Id : 640653906844 Question Type : SA Calculator : None

Correct Marks : 1

Question Label : Short Answer Question

Calculate the Economic Order Quantity (EOQ).

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

8000

Question Number : 127 Question Id : 640653906845 Question Type : SA Calculator : None

Correct Marks : 1

Question Label : Short Answer Question

Calculate the number of orders per year.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

10

Question Number : 128 Question Id : 640653906846 Question Type : SA Calculator : None

Correct Marks : 1

Question Label : Short Answer Question

Calculate the annual order cost.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

12000

Question Number : 129 Question Id : 640653906847 Question Type : SA Calculator : None

Correct Marks : 1

Question Label : Short Answer Question

Calculate the annual inventory carrying cost.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

12000

Sub-Section Number : 13

Sub-Section Id : 640653134759

Question Shuffling Allowed : No

Question Id : 640653906839 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Calculator : None

Question Numbers : (130 to 132)

Question Label : Comprehension

Suppose you want to invest \$4000 now and \$2000 at the start of years 2 to 4. The interest rate offered by First Bank is 8% compounded annually, and the bonuses over the next 4 years are 1.8%, 1.7%, 2.1% and 2.5%, respectively. The annual interest rate offered by Second Bank is 0.2% lower than that of First Bank, but its bonus is 0.5% higher. The objective is to maximize the accumulated capital at the end of 4 years. If you apply Dynamic Programming to this problem, answer the given subquestions:

Sub questions

Question Number : 130 Question Id : 640653906840 Question Type : MCQ Calculator : Yes

Correct Marks : 1

Question Label : Multiple Choice Question

How is a stage represented by?

Options :

6406533052428. ✓ Year

6406533052429. ✗ Capital available for investment at the start of a year

6406533052430. ✗ Amounts invested in First Bank

6406533052431. ✗ Amounts invested in First Bank and Second Bank

Question Number : 131 Question Id : 640653906841 Question Type : MCQ Calculator : Yes

Correct Marks : 1.5

Question Label : Multiple Choice Question

What are the alternatives in a stage?

Options :

6406533052432. ✖ Year

6406533052433. ✖ Capital available for investment at the start of a year

6406533052434. ✖ Amounts invested in First Bank in a year

6406533052435. ✔ Amounts invested in First Bank and Second Bank in a year

Question Number : 132 Question Id : 640653906842 Question Type : MCQ Calculator : Yes

Correct Marks : 2

Question Label : Multiple Choice Question

How state variable is defined for this problem?

Options :

6406533052436. ✖ Year

6406533052437. ✔ Capital available for investment at the start of a year

6406533052438. ✖ Amounts invested in First Bank in a year

6406533052439. ✖ Amounts invested in First Bank and Second Bank in a year

Sub-Section Number : 14

Sub-Section Id : 640653134760

Question Shuffling Allowed : No

Question Id : 640653906848 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Calculator : None

Question Numbers : (133 to 134)

Question Label : Comprehension

The owner of a newsstand wants to determine the number of newspapers of *USA* Now to be stocked at the start of each day. The owner pays 30 cents for a copy and sells it for 75 cents. The sale of the newspaper typically occurs between 7:00 and 8:00 am. Newspaper left at the end of the day are recycled for an income of 5 cents a copy. How many copies should the owner stock every morning, assuming that the demand for the day can be described as:

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 133 Question Id : 640653906849 Question Type : SA Calculator : None

Correct Marks : 2

Question Label : Short Answer Question

A normal distribution with mean 300 copies and standard deviation 20 copies.

[Standard_Norm_inv(critical fractile)=0.643]

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range